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NO I have not used such tools

Report

Topic modelling Pipeline for research Abstract

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Introduction

This pipeline is created so that it can process raw, unstructured text and able extract meaningful full insight and identify trends with the given dataset perform topic modelling with Latent Dirichlet Allocation (LDA). The pipeline that is created contains comprehensive preprocessing, advance statistical and visual analysis and scalable architecture. I have used Apprenticeship pipeline for development and validation but the pipeline work effectively with other datasets. I have run the pipeline with **genAI dataset** and it perform well. In this report I have detailed all steps for using the pipeline and thoroughly described its feature.

Important instruction for running the pipeline are highlighted in red. Below in the pipeline running instruction section.

1.Pipeline usage instructions

In this there are instruction to explain how to setup and run the pipeline.

1.1 Directory structure

dpams_pipeline/

- |— **data/**
 - | |— **apprenticeship_abstracts.csv**
 - | |— **genAI_abstracts.csv**
- |— **docs/**
 - | |— **README.md**
 - | |— **dmaps.pdf**
- |— **output/**
 - | |— **heatmap.png**
 - | |— **statistics.csv**
 - | |— **top_terms_plot.png**
 - | |— **wordcloud_Topic_1.png**
 - | |— **wordcloud_Topic_2.png**
- |— **scripts/**
 - | |— **1. Installing packages.R**
 - | |— **2. Preprocessing.R**
 - | |— **3. LDA_modelling.R**
 - | |— **4. Visualization.R**
 - | |— **5. Variance and standard deviation**
- |— **_pipeline.Rproj**
- |— **pipeline.R**
- |— **.gitignore**

The input dataset `apprenticeship_abstracts.csv` and `gen_AI.csv` is located in the ***data/*** folder of the directory.

1.2 Commands to run

Open RStudio and load *dpmas_pipeline.Rproj* file

Run *pipeline.R* or it could be run manually in a sequential order.

1.3 Main input requirements

The CSV file stored in /data which contains apprenticeship_abstracts.csv and gen_AI.csv .

The pipeline.R file which contains the main script.

1.4 Output Explanation

Cleaned Data: the clean abstract is saved in the **clean_data.csv** in the dataset the preprocessing steps.

Visualisation: PNG's saved in the /output folder lie Top_term_plot.png, word_cloud_1.png , heatmap.png etc

Statistics: A CSV file named *statistics.csv* which is generated contains variance and standard deviation for document length and term frequency.

Top topic: the CSV file named topic contain top 10 each topic.

1.5 Pipeline running instructions

1. Make sure **R** and **Rstudio** is installed in you system
2. In the directory open the **dmaps_pipeline.Rstudio** to run script
3. The main directory named **dmap_pipeline/** contains all the necessary files that are necessary. That are required to run this pipeline effectively including the Readme file. As shown in the directory structure above.
4. **Scripts**- contain all the necessary scripts
Data – it contain both the dataset both of them could be used to run pipeline.
Output- it contain all output files like PNG's and CSV's.
Readme- readme.md with description of the directory and instruction on how to run.

To start install all the necessary libraries present in **1. Installing packages.R** file.

```
# Required libraries

install.packages("tidytext")
install.packages("tidyverse")
install.packages("topicmodels")
install.packages("ggplot2")
install.packages("dplyr")
install.packages("tidyr")
install.packages("tibble")
install.packages("tm")
install.packages("wordcloud")
install.packages("conflicted")

library(tidytext)
library(tidyverse)
library(topicmodels)
library(ggplot2)
library(dplyr)
```

```
library(tm)
library(wordcloud)
library(pheatmap)
library(SnowballC)
library(powerLaw)
```

You can run scripts one by one (recommended) Which is divided into four phases. And kept into four folders with their name on it **1.installing packages.R**, **2. Preprocessing.R**, **3. LDA_modelling**, **4.Visualisation.R** **5. Variance and standard deviation.R** Run the code with code inside the script folder in this order 1 to 5.

phase 1 Loading the Dataset

```
data <- read.csv("data/apprenticeship_abstracts.csv")
```

Data Loading snippet

Currently apprenticeship_abstracts.csv file is data loading code which code replace with GenAI.csv or any other csv file to perform

Important

You can also run the pipeline all at once by running the script inside *pipeline.r*

But for smooth operation it is recommend to install above mentioned packages first.

Visualisation

- It is highly recommended to run the visualisation one after the other to see the output in the plot.
- And for the word cloud you need to press **ENTER** to see all the five word cloud of each topic.

Other than this all the visualisation is saved in the output folder.

2. Detailed Pipeline overview (key four stages of the data pipeline)

The pipeline consists of four core stages:

1. Data Preprocessing
2. Statistical Analysis
3. Topic Modelling
4. Visualization with analysis

Here is further breakdown of all the key stages with explaining on why it is used along with the output.

2.1 Preprocessing

To clean and preprocess raw data making it suitable for modelling. We start with

1. Load dataset and Remove empty the abstracts

Purpose : The loading of dataset is the foundation of every pipeline. Abstracts are removed to ensure the quality also ensures that analysis only includes the document that have content.

Code snippet

```
# Loading dataset

data <- read.csv("G:/My Drive/pipeline/apprenticeship_abstracts.csv")

data <- data[!is.na(data$Abstract) & data$Abstract != "", ]
```

Code Snippet

```
cat("Number of rows before removing empty abstracts:", nrow(data), "\n")
Number of rows before removing empty abstracts: 1000
```

After

```
cat("Number of rows after removing empty abstracts:", nrow(data), "\n")
Number of rows after removing empty abstracts: 946
```

2. Remove custom words and defining whitelist

Purpose: To remove irrelevant words. To the focus of the analysis is on meaningful and unique data.

Code snippet

```
# Defining Custom Stop Words
```

```
custom_stop_words <- c("apprenticeship", "study", "research", "analysis",  
"project",  
"training", "education", "students", "learning",  
"skills")
```

```
# Define Whitelist for Critical Terms to Preserve
```

```
whitelist <- c("education", "indigenous-centric", "apprenticeship")
```

Snippets

Before

The study emphasizes indigenous-centric education.

After

emphasizes indigenous-centric

.

3. Preprocessing text and stemming

Purpose: Converting to lowercase, Removing punctuation, extra space, trim whitespace. *Stemming* It's a technique to simplify the words to its base form therefore improving the analysis

Code snippet

```
# Preprocessing text
```

```
data$cleaned_abstract <- tolower(data$Abstract) # Convert to lowercase  
data$cleaned_abstract <- gsub("[^a-z\\s]", "", data$cleaned_abstract, perl =  
TRUE) # Remove punctuation  
data$cleaned_abstract <- gsub("\\s+", " ", data$cleaned_abstract, perl =  
TRUE) # Remove extra spaces  
data$cleaned_abstract <- trimws(data$cleaned_abstract) # Trim whitespace  
  
# stemming  
  
data$cleaned_abstract <- wordStem(data$cleaned_abstract, language = "en")
```

Snippets

Before

With a focus on the Western-Canadian province of Manitoba, this article aims to develop a conceptual and empirical exploration of how apprenticeship learning can be transformed to meet the needs of Indigenous apprentices.

After

focus westerncanadian province manitoba article aims develop conceptual
empirical exploration can transformed meet needs indigenous apprentices

4. De-duplication

Purpose: To ensure that each entry is unique by removing repeated data. As duplicated extract can skew the analysis.

Code snippet

```
# De-duplicating Entries  
  
data <- data %>% distinct(Abstract, .keep_all = TRUE)
```

Output:

Snippets

	Metric	Count
1	Rows Before De-duplication	1000
2	Rows After De-duplication	947

2.2 Statistical analysis (insight and trends)

After preprocessing some statistics were examined to understand the distribution of the dataset. Variance and standard deviation and term frequencies are calculated

1. Document length Statistics

Analysing the term frequencies and document length

Purpose: to help identify most common words in the dataset highlighting key themes or pattern for further analysis.

Compute summary statistics (variance and standard deviation)
Helps measure variability of the document length and term to provide insight into diversity of dataset.

Code snippet

```
# Variance and Standard Deviation of Term Frequencies  
  
term_frequencies <- colSums(as.matrix(dtm))  
term_stats <- data.frame(  
  Metric = c("Variance", "Standard Deviation"),  
  Type = "Term_Frequency",  
  Value = c(var(term_frequencies), sd(term_frequencies))  
)
```

Code for calculating term frequency


```
# Variance and Standard Deviation of Document Lengths
```

```
doc_lengths <- rowSums(as.matrix(dtm))  
doc_stats <- data.frame(  
  Metric = c("Variance", "Standard Deviation"),  
  Type = "Document_Length",  
  Value = c(var(doc_lengths), sd(doc_lengths))  
)
```

Code for calculating document length

Output: *statistics.csv* summarizing document length and term frequency also table below showing the output.

Category	Value
Term Frequency Variance	2781.009
Term Frequency Standard Deviation	52.73527
Document Length Variance	1221.222
Document Length Standard Deviation	34.94599

Variance and Standard deviation Metrics of apprenticeship_abstracts

These values are reasonable and typical for dataset research abstracts as well as textual summary. Low variance to term frequency indicates. That dataset is relatively uniform in document length. Which is good for topic modelling.

2.3 Topic modelling

The purpose is to quantifying so that algorithm can process it. Each row in this represents a document and each column represents a term.

Steps:

1. Create a Document term matrix for (DTM)

The matrix entries represent term frequency of terms across documents. It quantifies text data into numeric form so algorithm can process it. Each row there is represents a document and each column a term.

Code snippet

```
# Create Text Corpus and Document-Term Matrix (DTM)

corpus <- Corpus(VectorSource(data$cleaned_abstract))
dtm <- DocumentTermMatrix(corpus)
dtm <- removeSparseTerms(dtm, 0.99) # this removes the sparse terms
```

DTM dimensions before removing sparse terms: 946 documents and 12744 terms	DTM dimensions after removing sparse terms: 946 documents and 1829 terms
--	--

This is test after preprocessing and tokenization with cleaned text

2. Fit the LDA model

Purpose: Applying the LDA algorithm to the data to train the model to discover hidden in the text by analysing word pattern and assigning probability to each topic.

Code snippet

```
k <- 5 # Number of topics
lda_model <- LDA(dtm, k = k, control = list(seed = 123))
```

Explanation of the *K* it is assumed that there are 5 dominant topics in the dataset. If we adjust the value of *K* it can yield different results.

3. Extracting top terms from each topic

For Identifying the most significant word related to each topic. It helps defining the themes or concept.

Code snippet

```
beta <- posterior(lda_model)$terms # Extracting Term.
beta <- as.data.frame(beta)
colnames(beta) <- make.names(colnames(beta), unique = TRUE)
rownames(beta) <- paste0("Topic_", seq_len(nrow(beta)))
#Ensure Beta Matrix is a Data Frame with Unique Column Names
```

The Beta Matrix show probability of each term belonging to each topic.

4. Visualizing the top term topics

Purpose: To identify the top 10 of this topic to interpret what each topic represents.

Code snippet

```
# Converting Beta Matrix to Long Format

topic_terms_long <- beta %>%
  rownames_to_column("topic_label") %>%
  pivot_longer(cols = -topic_label, names_to = "term", values_to = "beta")
```

5. Grouped **Bar plot** to visualise to top 10 terms of each topic

Code snippet

```
# Step 13: Select Top 10 Terms per Topic
top_terms <- topic_terms_long %>%
  group_by(topic_label) %>%
  slice_max(beta, n = 10) %>%
  ungroup() %>%
  mutate(term = reorder_within(term, beta, topic_label))

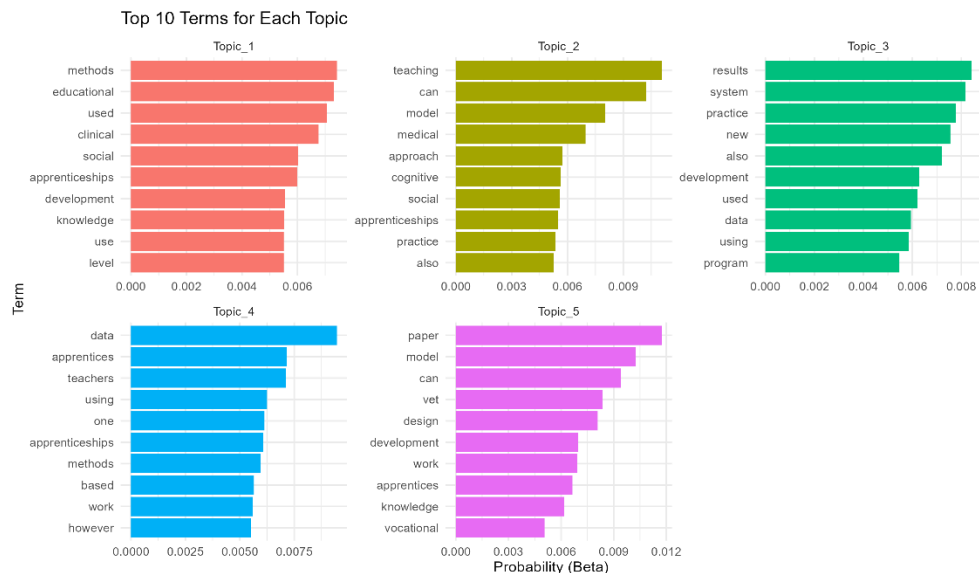
# Step 14: Visualize Top Terms per Topic
top_terms_plot <- ggplot(top_terms, aes(beta, term, fill = topic_label)) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ topic_label, scales = "free") +
  scale_y_reordered() +
  labs(title = "Top 10 Terms for Each Topic",
       x =
       y = "Term") +
  theme_minimal()
```

The resulting file it shows the top ten term each of the 5 topics. Which helps in labelling the topics.

Output:

topic_terms_long.csv saved as CSV file in output folder of the directory. Its top term per topic with the probabilities.

Visual representation (also saved in output folder)



2.4 Visualisation and analysis

This is section to interpret results visually making pattern trends clear.

Steps

1. **Generate tools like word clouds, power-law distribution plots and heatmap.** To visually summarize data, highlighting key pattern and make relationship that complex easy to understand.

Code snippet

Word cloud

```
# Generating Word Clouds for Topics

for (t in unique(topic_terms_long$topic_label)) {
  # Filtering terms for the current topic

  topic_words <- topic_terms_long %>% filter(topic_label == t)

  # Skip the topic if no terms or all term probabilities are zero

  if (nrow(topic_words) == 0 || all(topic_words$beta == 0)) {
    cat("Skipping", t, "- No terms with nonzero frequencies.\n")
    next
  }
}
```

```

}

wordcloud(words = topic_words$term, freq = topic_words$beta,
          max.words = 50, random.order = FALSE)

# Confirmation message

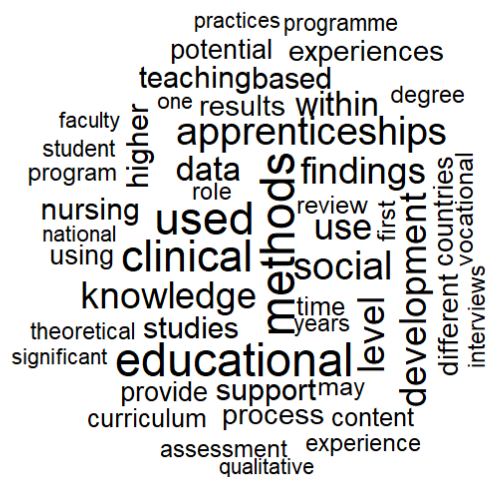
cat("Word cloud displayed for", t, "\n")

# Pause to allow viewing each word cloud

readline(prompt = "Press [Enter] to view the next word cloud...")
}

```

Output:



heatmap

```

# Filter the top 100 terms with that have the highest variance

top_terms <- apply(beta, 2, var) %>% sort(decreasing = TRUE) %>% head(100)
filtered_beta <- beta[, names(top_terms)]

pheatmap(filtered_beta,
          cluster_rows = TRUE,
          cluster_cols = TRUE,

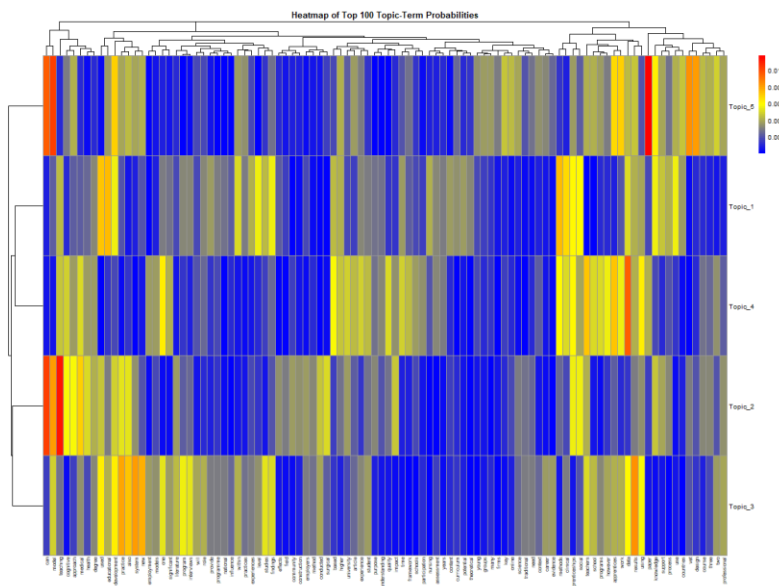
```

```

main = "Heatmap of Top 100 Topic-Term Probabilities",
fontsize_row = 10,
fontsize_col = 8,
angle_col = 90,
show_rownames = TRUE,
show_colnames = TRUE,
color = colorRampPalette(c("blue", "yellow", "red"))(50))
# Divergent color scheme

```

Output:



This heatmap is illustrates the top 100 term across the 5 topics.

- The horizontal clusters divide the term into clear group, illustrates that how particular term associates with a particular topics.
- The yellow red or red represents high probabilities, also the term that are associated with particular topics. And blue represents low probabilities it is indicating the term with low relevance to the topic.

Variations with in the topics

Topic 1,2 and 4 appears more populated with high probability terms hence well-designed topic theme.

Whereas Topic 5 less focus broader topic.

Keys Insight

- This analysis helps to the most defining term per topic also distinction among topic.
- It highlights how the unique term are distributed across.

This heat map is affective in highlighting term contribution per topic.

3.Evidence and Justification of additional features

This s the explanations, evidence and justification of all the additional features.

3.1Custom words and stemming

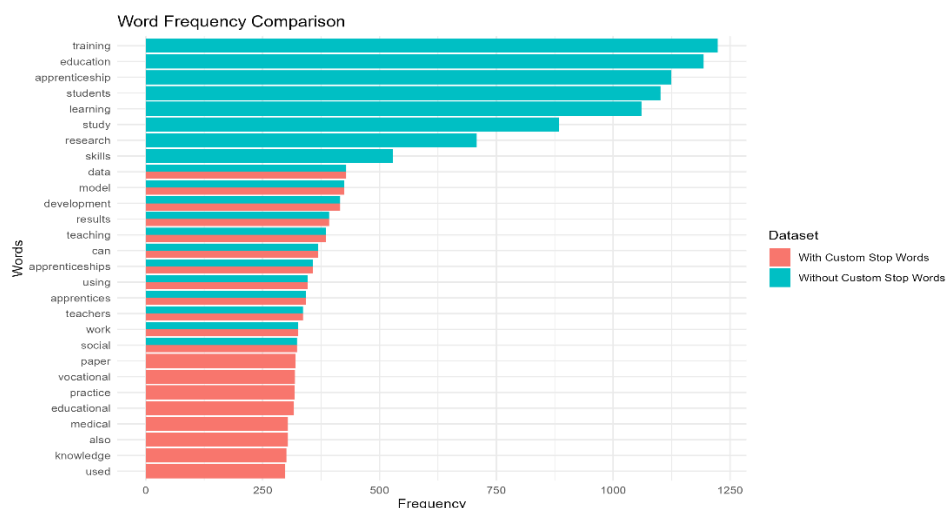
The main purpose of using custom words and stemming is the pipeline is to refine input data by removing an irrelevant term and reducing the to their base form. it improved the focus and effectiveness of topic modelling.

Before this frequent term included redundant terms like education and training.

Evidence:

6. table

Before cleaning	After cleaning
Apprenticeships, Education, training	Apprenticeships, development, Model



3.2 De- duplication

In large datasets certain abstracts appears multiple times which might skew the frequency and topic distribution. It is essential in pipeline for removing the duplicate entries hence ensures the reliability and accuracy of the analysis.

Evidence:

7. Table

<i>Metric Count</i>	Count
1. Total Words Before	12909
2. Total Words After	12899
3. Duplicates Removed	10
4. Duplicates Retained	12899

8. CSV file duplicates.csv

3.3 Stop words and Whitelist

It helps streamline the dataset by excluding the common words that are not informative. Therefore, including only relevant words. Which enhance the precision of the analysis. While the Whitelist preserve crucial context e.g. apprenticeship.

evidence

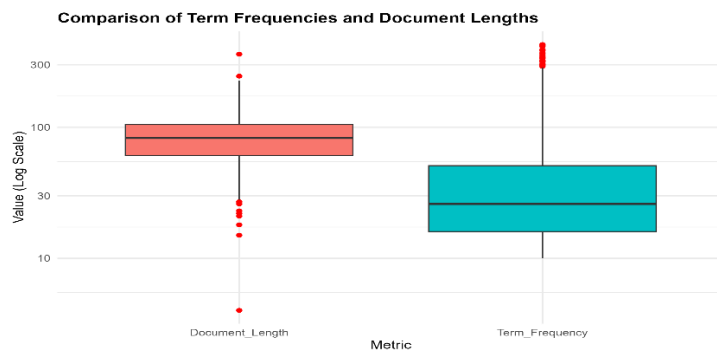
Before cleaning	After cleaning
"Apprenticeship has been a cornerstone of skill development..."	"apprenticeship cornerstone skill development..."

1. Statistical outputs

Analysing the variance and standard deviation of document length it helps us to gauge the consistency of the data distribution. Which reveal whether the data exhibit substantial diversity or maintains uniformity in content depth and word usage.

Evidence: Table

Category	Value
Term Frequency Variance	2781.009
Term Frequency Standard Deviation	52.73527
Document Length Variance	1221.222
Document Length Standard Deviation	34.94599



Document Length variance and standard deviation: Represented by red box plot.

Term frequency variance and standard deviation: Represented by teal box plot.

4. Analysis and insight

Discussion of Topic Model

Overview

In this pipeline LDA is applied to identifying latent topic with text data discussion. The model is set on a $K=5$ which is based on the dataset size and the diversity of the content.

4.1 Summary of the key terms and their interpretation

Topic summary:

Topic 1: Key term from the first topic are development knowledge, education, methods, clinical etc.

Interpretation: Integration of knowledge development and educational technique in clinical settings.

Topic 2: Key term from the Second topic are teaching, apprenticeships, cognitive, model, social etc.

Interpretation: The role of cognitive and social frameworks in educational practices in apprenticeships.

Topic 3: Key term from the third topic are development system, development, practice, program, results etc.

Interpretation: The creation and implementation of the development system with emphasis on evaluation their output.

Topic 4: Key term from the fourth topic are development data, teacher, method, based, work ,however etc.

Interpretation: How the data influences the approach of the teacher , training and personal growth.

Topic 5: Key term from the fifth topic are development vocational, design, vet, paper ,knowledge etc.

Interpretation: Applied knowledge on vocational education and training with innovative design

Evaluation of the topics

9. Each topic covers about unique theme which indicates a minimum overlap between the topics.
10. The consistency between the topics and the key term indicates that the model effectively captures patterns and it organizes thematic elements.

4.2 Statistical trend

1. Document length variation: There is a high variance of standard deviation which indicated that some document are significantly longer then other.

2. Term frequency spread: If few term that are dominant than others a power law or the frequency histogram could indicate that certain words are appearing dis proportionality.

4.3 Visual analysis

1. Word cloud: It show the most common words term in the form of collage of words.

2. Heatmap: It shows which terms are most prevalent in the topic.

5. Conclusion

The pipeline here presents demonstration a well-structured well way to handle raw unstructured text data. A good structured is followed across from the initial loading phase to the preprocessing, topic modelling and visualisation. Removing irrelevant customs words, duplication words and to the whitelist, stemming and tokenization. All these ensures that the further analysis and topic modelling represents the true nature of the data. The LDA represents interesting patterns.

Keys takeaway:

The importance of preprocessing: as it is significant in the analysis of the data further.

LDA: Gives a topic insight K=5 reveals meaningful structure, in the dataset, which unveils core themes.

Visualisation: the has huge for the right interpretation of the pipeline plots like bar, charts, heatmaps, power law distribution etc which provides useful key insight which could be overlooked by numeric summarization.

6. Code Reference

Data Visualization:

- ggplot2 Book by Hadley Wickham: [ggplot2: Elegant Graphics for Data Analysis](#)

R Programming Language:

- Official CRAN Documentation: <https://cran.r-project.org/>

Topic Modeling Tutorials:

- Comprehensive Guide: <https://cran.r-project.org/web/packages/topicmodels/vignettes/topicmodels.pdf>